Quantitative Research Methods

Chapter 11: Survival Analysis and Censored Data

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HSBI

The course at a glance

	Lecture	Topic
	1	Ch. 1 + 2 — Introduction; what is statistical learning
	2	Ch. 2 — Accuracy, bias–variance, Bayes classifier, KNN
	3–4	Ch. 3 — Linear regression (estimation, inference, diagnostics)
	5–6	Ch. 4 — Classification (logistic regression, confusion matrix, ROC)
	7	Ch. 5 — Resampling (validation, CV, bootstrap)
	8	Ch. 6 — Model selection and regularisation (ridge, lasso)
	9	Ch. 7 — Moving beyond linearity (splines, GAMs)
	10	Ch. 8 — Tree-based methods (trees, forests, boosting)
	11	Ch. 9 — Support vector machines
	12	Ch. 10 — Deep learning (MLPs, CNNs, training)
▶	**13**	**Ch. 11 — Survival analysis and censored data**
	14	Ch. 12 — Unsupervised learning (PCA, clustering)
	15	Ch. 13 — Multiple testing (FWER, FDR)

Fifteen lectures of 180 minutes, after a taught precourse session. ▶ marks today's chapter. Short exercises every $\sim$20 min; extended exercises every $\sim$45 min; each chapter has a companion lab notebook.

Contents

Optional and advanced material is collected in the **appendix** at the end of the deck.

Notation in this chapter

Symbol	Meaning
T	the true event time — what we want, but often do not see
C	the censoring time (end of study, drop-out, last contact)
$Y = \min(T, C)$	the *observed* time for a subject
$\delta = 1\{T \leq C\}$	event indicator: 1 if the event was seen, 0 if censored
$S(t) = \Pr(T > t)$	survival function: chance of surviving beyond t
$h(t)$	hazard: instantaneous event rate at t among those still at risk
$H(t) = \int_0^t h(u)\,du$	cumulative hazard, with $S(t) = e^{-H(t)}$
$t_1 < \cdots < t_K$	the distinct times at which events are observed
d_k	number of events at t_k
r_k	number *at risk* just before t_k (alive and still followed)
$\hat{S}(t)$	the Kaplan–Meier (product-limit) estimator
O_1, E_1, V_1	observed, expected and variance of group-1 events (log-rank)
$x_i,\ \beta$	covariate vector of subject i; vector of log hazard ratios
$h_0(t)$	baseline hazard — the part the Cox model never estimates
$R(t)$	risk set at t: $\{i : Y_i \geq t\}$
$\mathrm{PL}(\beta)$	Cox partial likelihood
$\mathrm{HR} = e^{\beta_j}$	hazard ratio for a one-unit increase in x_j

ISLP writes the distinct event times $d_1 < \cdots < d_K$ with q_k events and r_k at risk; we use t_k, d_k, r_k to keep δ free for the event indicator.

Sources and references

Primary textbook

James, G., Witten, D., Hastie, T., Tibshirani, R., & Taylor, J. (2023). *An Introduction to Statistical Learning, with Applications in Python.* Springer.

Data used in this chapter

BrainCancer — 88 patients treated with stereotactic radiation, followed for up to 82.6 months; 35 deaths and 53 censored observations. **Publication** — time from completion to publication for 244 clinical trials funded by the US NHLBI. Both ship with the course as CSV files.

Learning objectives

By the end of this chapter you will be able to:

- **Explain** why censored outcomes break ordinary regression and classification, and what right-censoring means formally.
- **State** the independent-censoring assumption and recognise settings in which it fails.
- **Compute** a Kaplan–Meier curve by hand and read a median, a survival probability and a confidence band off it.
- **Apply** the log-rank test to compare two survival curves, and say what it cannot detect.
- **Fit and interpret** a Cox proportional-hazards model, reading $e^{\hat{\beta}_j}$ as a hazard ratio.
- **Check** the proportional-hazards assumption and **choose** a remedy when it fails.

Roadmap of this chapter

Learning from data you have only partly seen

1. The problem: the response is a *time*, and for many subjects the clock is still running.
2. Censoring, stated properly — and the one assumption everything rests on.
3. The survival curve and its Kaplan–Meier estimate.
4. Comparing two curves: the log-rank test.
5. Adding covariates: the Cox proportional-hazards model.
6. Practice: checking proportional hazards, and what to do when it fails.

Where this chapter is used in industry

Sector	Concrete application	Which decision it drives
Pharma, medtech	Overall and progression-free survival in oncology trials	Whether the drug goes to the regulator at all
Subscription tech	Time to churn for customers who have *not yet* churned	Retention spend, and which cohorts get it
Banking, credit	Time to default or to early repayment on a loan book	Provisioning, pricing, and prepayment risk
Insurance	Time to lapse of a policy; time to claim	Reserving and commission clawback rules
Manufacturing	Time to failure of a component still in the field	Warranty terms and maintenance intervals
HR analytics	Time to resignation among current employees	Which retention interventions are funded

In industry — The common thread

In every row the interesting subjects are the ones *nothing has happened to yet*. A churn model that quietly drops all still-active customers is trained on the unhappy minority; a warranty analysis that ignores components still running will promise a shorter life than the product has.

Industry case in depth: a subscription business models churn

In industry — The data as it actually arrives

A SaaS firm has 180 months of history. For a customer who cancelled in month 14 the response is unambiguous: $T = 14$. For a customer who signed up eight months ago and is still paying, all we know is $T > 8$ — a **right-censored** observation. At any snapshot the majority of the book is censored, and the newest, fastest-growing cohorts are censored the most heavily.

Why the naive pipeline fails — and what replaces it

The tempting shortcut is a classifier for "churned in the last 12 months, yes/no". It throws away *when*, cannot use customers with less than 12 months of tenure, and silently changes definition every time the snapshot moves. The survival framing keeps every customer: Kaplan–Meier for the cohort curve, Cox for the drivers, and the fitted curve integrates into an expected customer lifetime.

Who signs off — and the honest caveat

Finance owns the resulting lifetime-value number, so the curve has to be defensible. The caveat: customers who leave *because* a price rise is coming are not censored independently of their risk — and no amount of estimation fixes that.

Outline

The response is a time, and part of it is missing

We observe subjects from a starting point and wait for an **event**: death, churn, default, failure, publication.

Three questions of exactly this shape

- *What fraction of these patients will still be alive in two years?*
- *Do men and women in this cohort have the same survival?*
- *Holding tumour type fixed, how much does a ten-point drop in the Karnofsky score raise the death rate?*

Takeaway

The complication is not the response's *type* — times are just positive numbers — but that for many subjects the event **has not happened yet**. Their true time T is unknown; we only know it exceeds what we have watched.

Why the tools we already have fail

Naive framing	What it does with a subject still at risk	Why the answer is wrong
Linear regression of Y on x	treats the last-seen time as the event time	every censored Y is *too small*; the fit is biased downwards
Classification "event by $t^\star$?"	needs a label it does not have for anyone followed less than $t^\star$	discards recent subjects; the label changes whenever the snapshot moves
Mean of the observed Y_i	averages true times with truncated ones	estimates nothing in particular

Takeaway

Censoring is **not** missing data that we may drop or impute casually. Each censored subject carries real information — "survived at least this long" — and survival analysis is the machinery for spending it.

BrainCancer: the data we will use all chapter

Variable	Meaning	Values
`time`	months of follow-up, Y_i	0.07 to 82.56
`status`	event indicator δ_i	$1 =$ died (35), $0 =$ censored (53)
`sex`	patient sex	Female 45, Male 43
`diagnosis`	tumour type	Meningioma 42, HG glioma 22, Other 14, LG glioma 9
`ki`	Karnofsky performance index	40–100; higher is healthier
`gtv`	gross tumour volume, cm^3	0.01–34.64
`loc`	tumour location	Supratentorial 69, Infratentorial 19
`stereo`	radiation method	SRT 65, SRS 23

Establish the coding before anything else

We checked the coding against the data, not against memory: with $\delta = 1$ read as *death* the sex log-rank statistic is 1.44, exactly ISLP's published value. Read the other way round every curve in the chapter inverts silently. 60.2% of these patients are censored — one missing `diagnosis` leaves 87 complete cases for the Cox fit.

Three tempting shortcuts, and the damage each does

Median survival on BrainCancer — four ways

(a) **Drop the** 53 **censored patients.** The 35 remaining deaths have median 11.6 months. But dropping the censored keeps only patients who died *during* the study — the short-lived ones — so survival is far **understated**.

(b) **Call the censoring time a death.** All 88 "events" give median 23.7 months. Every censored Y_i is a lower bound treated as exact, so this still **understates** survival.

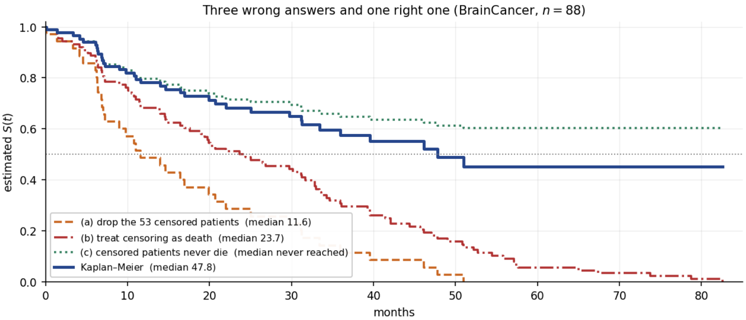
(c) **Assume the censored never die.** Then $\hat{S}(24) = 1 - 25/88 = 0.716$ and the curve flattens at $1 - 35/88 = 0.602$: the median is *never reached*. Survival is **overstated**.

$\Rightarrow$ **Kaplan–Meier** uses all 88 patients honestly: median 47.8 months, with $\hat{S}(24) = 0.683$.

Takeaway

Two of the shortcuts halve the answer and one inflates it. The errors do not cancel, and none of them get smaller with more data.

The same four estimates, drawn



Takeaway

The Kaplan–Meier curve (thick blue) crosses 0.5 at 47.8 months. Shortcut (a) crosses at 11.6 and (b) at 23.7; shortcut (c) never gets below 0.602. Same 88 patients, four different stories.

Exercise 11.1 — Why the shortcuts fail [Concept]

Exercise

A streaming service has 10 000 subscribers. In the last year 1 200 cancelled; the rest are still subscribed, with tenures ranging from one week to nine years. An analyst wants “the average subscription length”.

1. The analyst averages the tenure of the 1 200 who cancelled. In which direction is the answer biased, and why?
2. A colleague instead averages current tenure over all 10 000. In which direction is *that* biased?
3. Which single piece of information does every still-subscribed customer contribute, and why is it not nothing?
4. Give one reason why a classifier for “cancelled within 12 months?” is a poor substitute.

Solution — Exercise 11.1

Solution

Method. Ask, for each proposal, *which subjects it keeps* and whether that selection is related to the outcome. Selection on the outcome is what creates the bias.

Part 1 — cancellers only: biased downwards. To have cancelled inside a one-year window you must have had a short subscription. Long-lived customers are systematically excluded, so the average is far too small — exactly shortcut (a), which gave 11.6 instead of 47.8 months on BrainCancer.

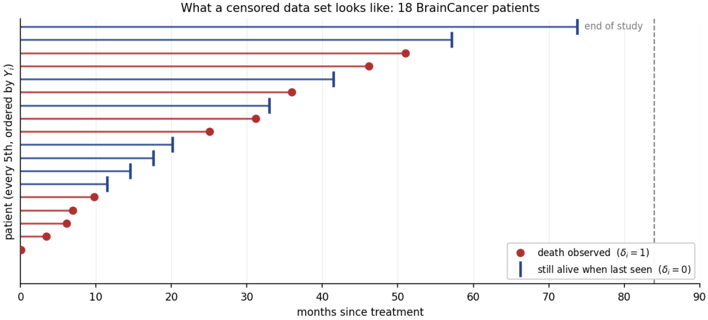
Part 2 — current tenure for everyone: also biased downwards. For an active customer, current tenure is a *lower bound* on the eventual subscription length. Averaging lower bounds as if they were the truth understates the mean — shortcut (b).

Part 3 — what a censored subject tells us. That $T_i > Y_i$. Concretely, an active customer of 30 months' tenure proves at least one customer survives past 30 months, and belongs in the *risk set* at every time up to 30. Kaplan–Meier and Cox both use exactly this.

Part 4 — against the classifier. Anyone who joined less than 12 months ago has no label at all, so the newest cohorts are silently dropped; and the target changes meaning every time the snapshot date moves.

Outline

The censoring picture — the figure to keep in your head



Takeaway

Each bar is one patient's follow-up. A **dot** ends a bar where a death was seen ($\delta_i = 1$): the time is exact. A **tick** ends a bar where the patient was last seen alive ($\delta_i = 0$): the true T_i lies somewhere to the *right* of the tick, and we will never know where.

What we actually observe

The observed pair

$$\boxed{Y_i = \min(T_i,\, C_i), \qquad \delta_i = 1\{T_i \le C_i\}.}$$

How to read this

- T_i is the **event time** we care about; C_i is the **censoring time** at which observation stops for reasons other than the event.
- We never see both. We see whichever comes first, Y_i, plus a flag δ_i saying which one it was.
- $\delta_i = 1$: the event happened at Y_i, so $T_i = Y_i$ exactly.
- $\delta_i = 0$: **right-censored** — all we learn is $T_i > Y_i$.
- A data set is therefore a list of pairs (Y_i, δ_i), plus covariates.

Takeaway

Every method in this chapter is a way of reading the likelihood of the pair (Y_i, δ_i) rather than of T_i.

The one assumption the whole chapter rests on

Independent (non-informative) censoring

$$\boxed{T_i \perp C_i \mid x_i}$$

Given the covariates, *when* a subject stops being observed carries no information about *when* their event would have occurred.

How to read this

- Equivalently: a subject censored at time t is representative of everyone still at risk at t. That is what lets us drop them from the risk set and carry on.
- It holds by construction for **administrative censoring** — the study ended, the extract was pulled — because the calendar does not know who is ill.
- It is an **assumption**, not a property of the data. No amount of data tests it: the counterfactual T_i for a censored subject is never observed.

Takeaway

Without it, none of Kaplan–Meier, log-rank or Cox estimates what we think it does. State it explicitly; then argue for it from how the data were collected.

When independent censoring fails

The failure to watch for

Patients drop out of a trial *because they are getting sicker* — too unwell to travel to the clinic. Their censoring times cluster just before the deaths that would have followed. Removing them from the risk set removes the highest-risk subjects, so every subsequent conditional survival is too high and $\hat{S}(t)$ is **optimistic**.

Three more from ordinary business data

- **Churn:** a customer whose card silently expires is recorded as active until the next billing cycle — censoring caused by disengagement.
- **Credit:** a borrower who refinances elsewhere is censored, and refinancing is chosen by the borrowers least likely to default.
- **Warranty:** a component returned by a customer who has stopped using the product is censored precisely because it was never stressed.

Takeaway

Ask of every censored record: *why did observation stop?* "The study ended" is safe. "The subject withdrew" needs a sensitivity analysis, and an honest sentence in the write-up.

Exercise 11.2 — Reading (Y, δ) off a study [Math]

Exercise

A study opens on 1 January 2020 and closes on 31 December 2022 (all times in *months from enrolment*). Four subjects:

	Enrolled	What happened
A	month 0	event at month 18
B	month 0	alive and in the study at closure
C	month 12	moved abroad at month 9 after enrolment
D	month 24	event at month 4 after enrolment

1. Give (Y_i, δ_i) for each subject.
2. Which censoring here is administrative, and which is not?
3. At time $t = 10$ months, which subjects are in the risk set $R(10)$?
4. Why is $\sum_i Y_i/4$ not an estimate of $E[T]$?

Solution — Exercise 11.2

Solution

Part 1 — the observed pairs. The study is open for 36 months, so a subject enrolled in month s can be followed for at most $36 - s$ months.

Subject	C_i	$Y_i = \min(T_i, C_i)$	δ_i	reason
A	36	18	1	event seen at 18
B	36	36	0	still event-free at closure
C	9	9	0	lost to follow-up at 9
D	12	4	1	event seen at 4

Part 2 — kind of censoring. B is **administrative**: the calendar ran out, which cannot depend on B's risk. C is **loss to follow-up**; independence is now an argument, not a fact — moving abroad is plausibly unrelated to the event, but it must be said out loud.

Part 3 — the risk set at $t = 10$. $R(10) = \{i : Y_i \geq 10\} = \{A, B\}$. Subject C left at 9 and D's event occurred at 4; both are gone. Note D contributes to $R(4)$ even though enrolled last — the clock is time *since enrolment*, not calendar time.

Part 4 — why the naive mean fails. $\frac{1}{4}(18 + 36 + 9 + 4) = 16.75$ mixes two exact times with two lower bounds. It is an average of $\min(T_i, C_i)$, which depends on the study design; enrol B a year earlier and the "estimate" changes although nothing about the disease did.

Outline

The object of interest: the survival curve

Definition

$\boxed{S(t) = \Pr(T > t)}$ a non-increasing function with $S(0) = 1$.

How to read this

- $S(t)$ answers the question a decision-maker actually asks: *what fraction is still event-free at t?* And $1 - S(t)$ is the cumulative incidence — the chance the event has happened by t.
- The **median survival** is the smallest t with $S(t) \leq 0.5$.
- S is a *whole function*: two groups can share a median and differ everywhere else, so we plot curves before testing anything.
- The **mean** $E[T] = \int_0^\infty S(t)dt$ needs the entire tail: on BrainCancer $\hat{S}$ stops at 0.452 after 82.56 months, so anything beyond is extrapolation. Use the **restricted** mean $\mathrm{RMST}(\tau) = \int_0^\tau \hat{S}(t)dt$ instead — 33.98 months for $\tau = 48$, 39.51 for $\tau = 60$.

Takeaway

Quote a median, a probability at a stated horizon, or an RMST *with* its horizon. A bare mean survival time from censored data is an extrapolation in disguise.

The idea: chain the conditional survivals

Surviving past t means surviving each earlier event time *in turn*:

Decomposition

$$S(t_k) = \Pr(T > t_k) = \prod_{j=1}^{k} \underbrace{\Pr(T > t_j \mid T > t_{j-1})}_{\text{survive step } j}$$

Why this is the whole trick

- Each factor is a *conditional* probability, estimable from the subjects who reached t_{j-1} — and censored subjects reached it too.
- So a subject censored at 30 months contributes to every factor up to month 30 and simply leaves the risk set afterwards.
- Estimate factor j by the obvious proportion: of the r_j at risk, d_j died, so $1 - d_j/r_j$ survived.

Takeaway

Censoring is handled by **shrinking the denominator**, not by deleting rows. That single move is what makes the estimator honest.

The Kaplan–Meier (product-limit) estimator

Rule

$$\boxed{\hat{S}(t) = \prod_{k\,:\,t_k \le t} \left(1 - \frac{d_k}{r_k}\right)}$$

How to read this

- $t_1 < \cdots < t_K$ are the distinct times at which **events** occur — censoring times never enter the product.
- d_k = events at t_k; $r_k = \#\{i : Y_i \geq t_k\}$ = subjects at risk just before t_k.
- $\hat{S}$ is a **step function**: flat between event times, dropping by the factor $1 - d_k/r_k$ at each t_k.
- Every censoring at time c removes one subject from all later r_k, so the *next* drop is bigger. That is how censored subjects influence the curve.
- With no censoring, r_k falls by exactly d_k each step and $\hat{S}$ collapses to the empirical survival function $\#\{Y_i > t\}/n$.

Takeaway

Non-parametric: no distribution for T, no model for the hazard, only the sequence of at-risk sets.

By hand: the first three deaths in BrainCancer

Three steps of the product

Ordered follow-up begins 0.07, 1.18^+, 1.41, 1.54^+, 2.03^+, 3.38, ... ($^+$ marks censoring).

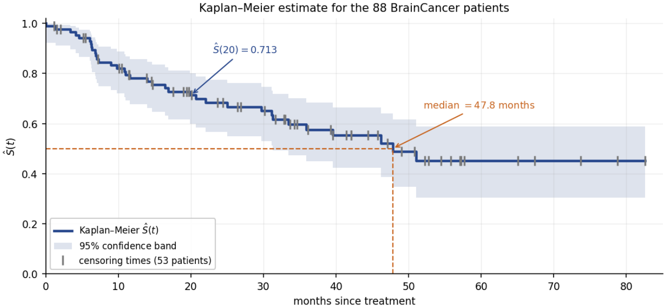
k	t_k	r_k	d_k	$\hat{S}(t_k)$
1	0.07	88	1	$1 \cdot \frac{87}{88} = 0.9886$
2	1.41	86	1	$0.9886 \cdot \frac{85}{86} = 0.9771$
3	3.38	83	1	$0.9771 \cdot \frac{82}{83} = 0.9654$

Note $r_2 = 86$, not 87: the censoring at 1.18 removed a subject before t_2. Between t_2 and t_3 two more censorings (1.54, 2.03) take r from 85 to 83 without the curve moving at all.

Takeaway

Censorings are invisible in the *numerator* and decisive in the **denominator**. Track r_k carefully and the rest is arithmetic.

The Kaplan–Meier curve with its confidence band



Takeaway

Read off: $\hat{S}(20) = 0.713$ with 95% interval [0.600, 0.800], and a median of 47.8 months. The band widens from left to right — only 48 of the 88 patients are still under observation after month 20 — and the curve flattens at 0.452 because the last observation is censored, so it never reaches zero.

Reading the curve without over-reading it

t (months)	10	20	40	60
$\hat{S}(t)$	0.821	0.713	0.552	0.452
95% interval	$[0.720, 0.888]$	$[0.600, 0.800]$	$[0.423, 0.664]$	$[0.305, 0.587]$

Three habits

- Quote $\hat{S}(t)$ *with* its interval; the right-hand end of a survival curve is always thin data.
- The median comes with a one-sided story here: its 95% interval is $[31.25, \infty)$, because $\hat{S}$ never gets far enough below 0.5 to pin down an upper limit.
- Never extrapolate past the largest observed time (82.56 months).

In industry — Insurance: the flat tail is a modelling decision

A lapse curve that flattens at 0.45 does not mean 45% of policies last forever — it means follow-up ran out. Reserving teams must either extrapolate with a parametric tail and say so, or report a restricted quantity such as $\mathrm{RMST}(\tau)$.

Censored observations do not step the curve down

Common mistake

Reading each tick mark as a small drop. A censoring at time c leaves $\hat{S}$ **unchanged** — nobody died. It only shrinks r_k for all $t_k > c$, so the *next* death causes a larger step. Many ticks in a row and then a big drop is the signature of thin data, not of a sudden hazard.

Two more honest readings

- $\hat{S}$ reaches 0 only if the *largest* observation is an event. Here it is censored, so the curve stops at 0.452 and the mean is not identified.
- If a group's curve never crosses 0.5, its median survival is "not reached" — report that, rather than the last value on the curve.

Takeaway

Always plot the censoring marks. A curve without them hides how much of its right-hand side rests on a handful of patients.

Exercise 11.3 — Kaplan–Meier by hand [Math]

Exercise

Six patients are followed (months, $^+$ = censored):

$$4, \quad 6^+, \quad 8, \quad 11^+, \quad 12, \quad 15.$$

1. Tabulate t_k, r_k, d_k for every *event* time.
2. Compute $\hat{S}(t)$ at each event time.
3. Give $\hat{S}(10)$ and the estimated median survival.
4. Redo part 2 pretending the two censored patients had died at 6 and 11. Compare.

Solution — Exercise 11.3

Solution

Parts 1–2 — the product. Event times are $4, 8, 12, 15$; r_k counts everyone with $Y_i \geq t_k$.

k	t_k	r_k	d_k	$\hat{S}(t_k)$
1	4	6	1	$1 \cdot (1 - \frac{1}{6}) = 0.8333$
2	8	4	1	$0.8333 \cdot (1 - \frac{1}{4}) = 0.6250$
3	12	2	1	$0.6250 \cdot (1 - \frac{1}{2}) = 0.3125$
4	15	1	1	$0.3125 \cdot (1 - \frac{1}{1}) = 0$

At $t_2 = 8$ the risk set is $\{8, 11^+, 12, 15\}$, so $r_2 = 4$: the patient censored at 6 has left, but was counted at t_1.

Part 3. $\hat{S}$ is flat between event times, so $\hat{S}(10) = \hat{S}(8) = 0.6250$. The median is the first t with $\hat{S}(t) \leq 0.5$, namely $t = 12$.

Part 4 — censoring treated as death. Now all six are events with $r = 6, 5, 4, 3, 2, 1$, giving $\hat{S} = \frac{5}{6}, \frac{4}{6}, \ldots$, i.e. the plain empirical curve 5/6, 4/6, 3/6, 2/6, 1/6, 0 at $4, 6, 8, 11, 12, 15$. The median falls to 11 and $\hat{S}(10) = 4/6 = 0.667$ — the curve is dragged down early and the median is understated, exactly shortcut (b) from the motivation.

Extended Exercise 11.1 — Kaplan–Meier in Python [Python]

Extended exercise (15 min)

Using `BrainCancer` and `lifelines`:

1. Fit the overall Kaplan–Meier curve. Report $\hat{S}(20)$ with its 95% interval and the median survival time.
2. Fit separate curves for men and women. Report each group's size, number of deaths and median.
3. Plot both curves with censoring marks, and say in one sentence what the plot suggests.
4. Verify by direct arithmetic that the first three steps of the overall curve are 0.9886, 0.9771, 0.9654.

Run this live

The worked solution sits in `chapter_11_lab.ipynb` under *Lecture exercises — worked Python solutions*: the cell headed *Extended Exercise 11.1 — Kaplan–Meier in Python [Python]*.

Solution — Extended Exercise 11.1 (1/2)

Solution — code

```
import pandas as pd, numpy as np # data and arithmetic
from lifelines import KaplanMeierFitter # product-limit estimator

BC = load('BrainCancer', index_col=0) # 88 x 8; status 1 = death
km = KaplanMeierFitter().fit(BC['time'], BC['status']) # censoring handled here
print('S(20) = %.4f' % km.survival_function_at_times(20).iloc[0])
print('95%% CI ', np.round(km.confidence_interval_.loc[:20].iloc[-1].values, 4))
print('median ', km.median_survival_time_) # first t with S(t) <= 0.5

for s in ['Female', 'Male']: # one curve per group
    m = BC['sex'] == s
    k = KaplanMeierFitter().fit(BC.loc[m, 'time'], BC.loc[m, 'status'])
    print(s, 'n=%d deaths=%d median=%.2f'
          % (m.sum(), BC.loc[m, 'status'].sum(), k.median_survival_time_))

et = km.event_table # r_k and d_k, tabulated
print(et.head(7)[['at_risk', 'observed', 'censored']]) # check the hand arithmetic
```

Solution — Extended Exercise 11.1 (2/2)

Solution — output and reading

Printout.

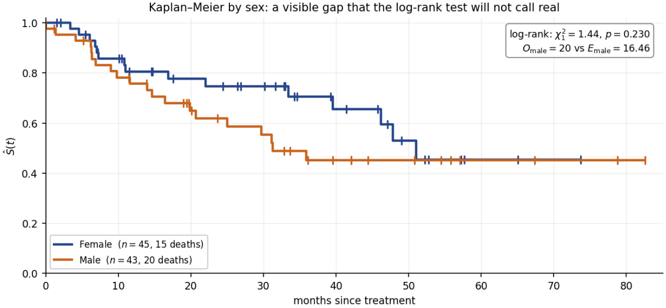
```
S(20) = 0.7132  95% CI [0.5996, 0.7998]  median 47.8
Female n=45 deaths=15 median=51.02    Male n=43 deaths=20 median=31.25
```

- **Part 1.** 71.3% of patients are still alive at 20 months; the interval $[0.600, 0.800]$ is already 20 percentage points wide with 88 patients — survival curves are data-hungry.
- **Part 2–3.** The male median is 31.25 months against 51.02 for women, and the male curve sits below the female curve almost everywhere. The plot *suggests* worse male survival; whether the gap is more than noise is the next section's question, and 45 vs 43 patients with 35 deaths in total is not much evidence.
- **Part 4.** `event_table` gives $r_k = 88, 86, 83$ and $d_k = 1, 1, 1$ at $t_k = 0.07, 1.41, 3.38$; the running product $\frac{87}{88} \cdot \frac{85}{86} \cdot \frac{82}{83}$ reproduces 0.9886, 0.9771, 0.9654.

Common mistake: passing `status` as a survival indicator rather than an event indicator. `lifelines` wants 1 = event; hand it `1−status` and every curve inverts while the code runs happily.

Outline

Two curves, one question



Takeaway

The gap looks real: medians 31.25 (men) against 51.02 months (women), and $\hat{S}(30) = 0.553$ against 0.747. But the bands overlap throughout. We need a test that compares the *whole curves* while respecting the censoring.

One 2×2 table at every death time

At time t_k	Died	Survived	At risk
Group 1	d_{1k}	$r_{1k} - d_{1k}$	r_{1k}
Group 2	d_{2k}	$r_{2k} - d_{2k}$	r_{2k}
Total	d_k	$r_k - d_k$	r_k

The null hypothesis table by table

Under $H_0 : S_1(t) = S_2(t)$ the d_k deaths at t_k are allocated purely by how many are at risk, so, conditioning on the margins,

$$E_{1k} = d_k \frac{r_{1k}}{r_k}, \qquad V_{1k} = \frac{d_k \,(r_{1k}/r_k)\,(1 - r_{1k}/r_k)\,(r_k - d_k)}{r_k - 1}.$$

(E_{1k} and V_{1k} are the hypergeometric mean and variance of d_{1k}.)

Takeaway

Censoring enters only through r_{1k} and r_{2k}, exactly as in Kaplan–Meier: a sequence of small fair-allocation checks, then one sum.

The log-rank statistic

Rule

Accumulate over all K death times, then standardise:

$$\boxed{Z = \frac{O_1 - E_1}{\sqrt{V_1}} = \frac{\sum_{k=1}^{K}(d_{1k} - E_{1k})}{\sqrt{\sum_{k=1}^{K} V_{1k}}}, \qquad Z^2 \overset{H_0}{\approx} \chi^2_1.}$$

How to read this

- $O_1 = \sum_k d_{1k}$: deaths actually seen in group 1; $E_1 = \sum_k E_{1k}$: the deaths group 1 would get if the groups were identical.
- $Z > 0$ means group 1 died *more* than its share of the risk sets warranted. Report Z^2 against χ^2_1, or Z against $N(0, 1)$ for a direction.
- Adding the tables, rather than pooling the data once, is what keeps the comparison valid under censoring.

Takeaway

The log-rank test is the observed-minus-expected count of events, scaled by its own standard error. Nothing is assumed about the shape of S.

BrainCancer by sex: the real numbers

Accumulating over all 35 death times

$$O_{\text{male}} = 20, \qquad E_{\text{male}} = 16.4605, \qquad V_{\text{male}} = 8.6969,$$

$$Z = \frac{20 - 16.4605}{\sqrt{8.6969}} = \frac{3.5395}{2.9491} = 1.200, \qquad Z^2 = \chi^2_1 = 1.4405, \qquad p = 0.230.$$

Men suffered 3.54 more deaths than their share of the risk sets predicts — a real but unremarkable excess, about 1.2 standard errors.

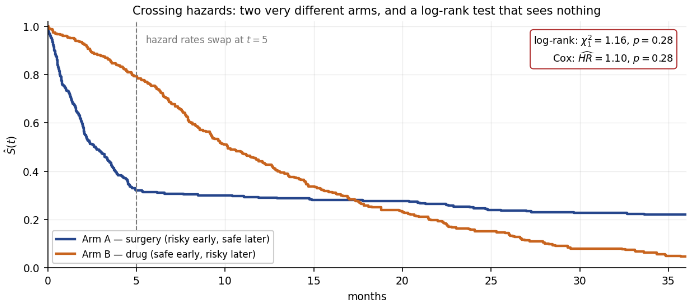
Takeaway

$p = 0.230$: with 88 patients and 35 deaths there is **no evidence** of a sex difference in survival. A visible gap between two curves and a 20-month gap between medians is entirely consistent with chance at this sample size.

In industry — Clinical development: powering on events rather than patients

The precision of a log-rank test is governed by V_1, which grows with the number of *events*. Trials are therefore sized in events, and event-driven analyses are triggered when the count is reached — recruiting more patients into a low-event setting buys surprisingly little power.

What the log-rank test cannot see



A large p-value is not "the arms are the same"

Simulated trial, 300 per arm: surgery is dangerous early and safe later, the drug is the reverse. At 6 months $\hat{S}_A = 0.313$ against $\hat{S}_B = 0.753$; at 36 months the ranking has reversed, 0.220 against 0.047. The log-rank statistic is $\chi^2_1 = 1.16$, $p = 0.28$: the early and late excesses **cancel in the sum** $O_1 - E_1$. Plot first, test second — and when curves cross, compare at fixed horizons or by RMST instead.

Exercise 11.4 — The log-rank test by hand [Math]

Exercise

Two treatment arms, times in months ($^+$ = censored):

$$\text{A: } 6,\ 7^+,\ 10,\ 15^+ \qquad\qquad \text{B: } 3,\ 5,\ 9,\ 12^+$$

1. List the death times and, at each, the numbers at risk r_{Ak}, r_{Bk}.
2. Compute O_A and $E_A = \sum_k d_k r_{Ak}/r_k$.
3. Given $V_A = 1.1893$, compute the log-rank statistic and compare it with χ^2_1 (the 5% critical value is 3.84).
4. State the conclusion, and one reason it should not surprise you.

Solution — Exercise 11.4

Solution

Parts 1–2 — the table. Death times are $3, 5, 6, 9, 10$ (censorings are not death times, but they do leave the risk sets):

t_k	r_{Ak}	r_{Bk}	r_k	d_k	d_{Ak}	$E_{Ak} = d_k r_{Ak}/r_k$
3	4	4	8	1	0	0.5000
5	4	3	7	1	0	0.5714
6	4	2	6	1	1	0.6667
9	2	2	4	1	0	0.5000
10	2	1	3	1	1	0.6667
					$O_A = 2$	$E_A = 2.9048$

At $t = 5$, arm A still has all four subjects at risk while B is down to $\{5, 9, 12\}$; by $t = 9$ arm A is $\{10, 15\}$ because 6 died and 7^+ was censored.

Part 3 — the statistic.

$$Z = \frac{2 - 2.9048}{\sqrt{1.1893}} = \frac{-0.9048}{1.0906} = -0.830, \qquad Z^2 = 0.688 < 3.84, \qquad p = 0.407.$$

Part 4 — conclusion. No evidence of a difference. Arm A had *fewer* deaths than expected ($Z < 0$), but with only five events the standard error swamps the gap: V_A grows with events, and five is not a study.

Common mistake

Building the tables at censoring times. Only death times contribute a term; a censoring merely shrinks r_{Ak} or r_{Bk} for later tables.

Outline

From two curves to many covariates: the hazard

Comparing curves group by group does not scale to `ki`, `gtv`, `diagnosis`, ... We need a regression — and it is built on the hazard.

Definition

$$\boxed{h(t) = \lim_{\Delta \to 0} \frac{\Pr\big(t < T \le t + \Delta \mid T > t\big)}{\Delta}}$$

How to read this

- A **rate**, not a probability: events per unit time *among those still at risk*. It can exceed 1 and has units 1/time.
- "Given that you have survived to t, how fast is the event arriving right now?" — the natural quantity for censored data, because it conditions on exactly the set we can still observe.
- Cumulative hazard $H(t) = \int_0^t h(u)du$ ties it back to survival: $S(t) = e^{-H(t)}$, and equivalently $h(t) = -\frac{d}{dt} \log S(t)$.
- So h, H and S are three views of one object: fix any one and the others follow.

Takeaway

Model the hazard and you have modelled the survival curve — with the censoring handled for free.

The Cox proportional-hazards model

Model

$$\boxed{h(t \mid x_i) = h_0(t)\,\exp\left(x_i^\top \beta\right) = h_0(t)\exp\left(\beta_1 x_{i1} + \cdots + \beta_p x_{ip}\right)}$$

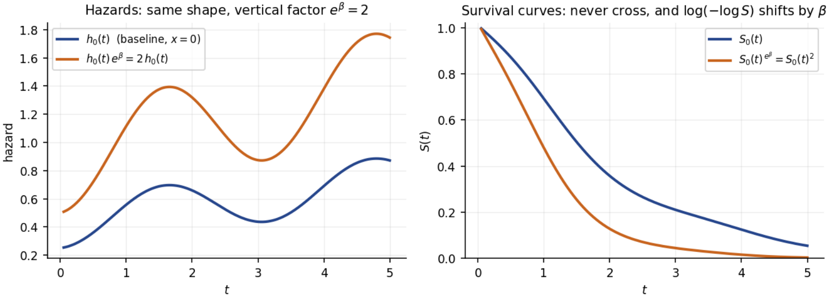
How to read this

- $h_0(t)$: the **baseline hazard**, shared by everyone and left completely unspecified — any non-negative function of time.
- $\exp(x_i^\top \beta)$: a subject-specific multiplier, constant in t and positive whatever β is. There is no intercept: it would be absorbed into h_0.
- **Semi-parametric**: non-parametric in time, parametric in the covariates.
- "Proportional": for two subjects the hazard *ratio* $\exp\{(x_i - x_j)^\top \beta\}$ does not depend on t — the curves may have any shape, but the vertical factor between them never changes.

Takeaway

All the flexibility goes into time, all the interpretation into β. That division of labour is why this model has dominated applied survival analysis since 1972.

What "proportional hazards" looks like



Takeaway

Left: the baseline hazard may wiggle as much as it likes; the covariate only multiplies it, here by $e^{\beta} = 2$. Right: the survival curves are then $S_0(t)$ and $S_0(t)^2$ — they **cannot cross**, and $\log(-\log S)$ shifts by the constant β. Both are testable consequences.

Estimating β without ever estimating h_0

The partial likelihood (Cox, 1972)

$$\boxed{\mathrm{PL}(\beta) = \prod_{i:\,\delta_i=1} \frac{\exp(x_i^\top \beta)}{\sum_{j\in R(Y_i)} \exp(x_j^\top \beta)}} \qquad R(Y_i) = \{j : Y_j \geq Y_i\}$$

Where the baseline went

- At each observed death ask: *among everyone still at risk, what was the chance that this subject was the one to die?* That chance is $h_0(Y_i)e^{x_i^\top \beta} / \sum_{j\in R(Y_i)} h_0(Y_i)e^{x_j^\top \beta}$, and $h_0(Y_i)$ is in every term of numerator and denominator, so it **cancels exactly**.
- Maximise $\log \mathrm{PL}$ numerically; $\hat{\beta}$ is consistent and asymptotically normal, with the usual Wald and likelihood-ratio tests. Censored subjects contribute no factor but populate the risk sets of earlier deaths.
- **Buys:** no distribution for T; only the *order* of the times matters. **Costs:** no absolute risk from $\hat{\beta}$ alone.

In industry — Credit risk: why ordering-only matters

A prepayment model fitted on loan age transfers to a book measured in payment counts, because the partial likelihood uses only who-was-at-risk-when.

Coefficients are log hazard ratios

Reading rule

$$\boxed{\frac{h(t \mid x_j + 1)}{h(t \mid x_j)} = e^{\beta_j} \quad \text{for every } t \qquad \Longrightarrow \qquad \beta_j = \log \mathrm{HR}_j.}$$

How to read this

- $e^{\beta_j} > 1$: higher x_j means a *higher* event rate, so shorter survival. $e^{\beta_j} < 1$: protective. $e^{\beta_j} = 1$: no effect.
- A c-unit change multiplies the hazard by $e^{c\beta_j}$ — choose c so the number is clinically meaningful (ten Karnofsky points, not one).
- Confidence intervals are computed on the β scale and then exponentiated, which is why hazard-ratio intervals are asymmetric around $e^{\hat{\beta}_j}$.
- Test $H_0 : \beta_j = 0$ (i.e. $\mathrm{HR} = 1$) with $z = \hat{\beta}_j / \mathrm{se}(\hat{\beta}_j)$.

Takeaway

One number per covariate, valid at all times — the payoff of the proportional-hazards assumption, and the reason it must be checked.

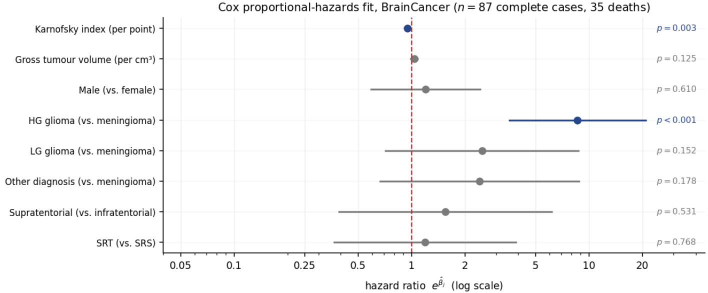
The fitted model on BrainCancer

Covariate	$\hat{\beta}$	se	$e^{\hat{\beta}}$	95% CI for $e^{\hat{\beta}}$	p
`ki` (per point)	−0.0550	0.0183	0.947	[0.913, 0.981]	0.003
`gtv` (per cm^3)	0.0343	0.0223	1.035	[0.991, 1.081]	0.125
`sex` Male	0.1837	0.3604	1.202	[0.593, 2.435]	0.610
`diagnosis` HG glioma	2.1546	0.4505	8.624	[3.566, 20.855]	< 0.001
`diagnosis` LG glioma	0.9150	0.6382	2.497	[0.715, 8.722]	0.152
`diagnosis` Other	0.8857	0.6579	2.425	[0.668, 8.803]	0.178
`loc` Supratentorial	0.4412	0.7037	1.555	[0.391, 6.174]	0.531
`stereo` SRT	0.1778	0.6016	1.195	[0.367, 3.884]	0.768

Housekeeping before interpretation

$n = 87$ complete cases, 35 deaths. `diagnosis` is coded against **Meningioma**; every diagnosis row is a comparison with that group. Overall likelihood-ratio test: $\chi^2_8 = 41.4$, $p = 1.8 \times 10^{-6}$ — the covariates together matter, even though most individually do not.

The same fit as a forest plot



Takeaway

The vertical line at $\mathrm{HR} = 1$ is "no effect"; intervals are plotted on a **log** scale, which is the scale on which they are symmetric. Only two intervals clear the line: `ki` (0.947, just left of it) and HG glioma (8.624, far right). Six covariates, 35 deaths — most intervals are simply wide.

Reading two coefficients out loud

Karnofsky index and a diagnosis

`ki`: $\hat{\beta} = -0.0550$, so one extra Karnofsky point multiplies the death rate by $e^{-0.0550} = 0.947$ — a 5.3% reduction. Per unit is too fine to discuss, so rescale:

$$e^{10\times(-0.0550)} = e^{-0.550} = 0.577.$$

"Holding tumour type, volume, site, sex and technique fixed, ten more Karnofsky points are associated with a 42% *lower death rate at every follow-up time."* **HG glioma vs. meningioma:** $e^{2.1546} = 8.62$, CI $[3.57, 20.86]$.

The death rate is 8.6 times higher, and even the optimistic end of the interval is a 3.6-fold increase. In survival terms at `ki` $= 80$: $\hat{S}(24) = 0.874$ for meningioma against 0.314 for HG glioma.

Takeaway

Always name the comparison, the units, and the phrase "holding the other covariates fixed". A hazard ratio without its reference level is not a result.

A hazard ratio is not a risk ratio

Three different quantities that students merge into one

- **Hazard ratio** e^{β}: ratio of *instantaneous rates* among those still at risk, at a given time. It is a ratio of *speeds*.
- **Risk ratio** $\frac{1-S_1(t)}{1-S_2(t)}$: ratio of cumulative probabilities by a *fixed* time t. It depends on t and is always closer to 1 than the hazard ratio.
- **Odds ratio** $\frac{p_1/(1-p_1)}{p_2/(1-p_2)}$: from logistic regression, on the odds scale, ignoring time entirely.

"$\mathrm{HR} = 8.6$" does **not** mean HG glioma patients are 8.6 times more likely to die, nor that they die 8.6 times sooner. At 24 months the actual risk ratio here is $(1 - 0.314)/(1 - 0.874) = 5.4$, and by 36 months it is 4.3 — the same constant hazard ratio produces a shrinking risk ratio.

Takeaway

Report a hazard ratio for the mechanism and a survival probability at a stated horizon for the decision. Never translate one into the other by intuition.

Exercise 11.5 — Hazard-ratio arithmetic [Math]

Exercise

A Cox model for time to churn is fitted with covariates `monthly_price` (in euros) and `onboarded` ($1 =$ completed onboarding):

$$\hat{\beta}_{\text{price}} = 0.024 \text{ (se } 0.006), \qquad \hat{\beta}_{\text{onboarded}} = -0.693 \text{ (se } 0.210).$$

1. Give the hazard ratio for a 10-euro price increase.
2. Give the hazard ratio for completing onboarding, and interpret it in one sentence.
3. Test $H_0 : \beta_{\text{onboarded}} = 0$ and give a 95% interval for its hazard ratio.
4. Two customers differ by 20 euros in price, and the cheaper one completed onboarding. What is their hazard ratio?

Solution — Exercise 11.5

Solution

Part 1 — rescaling. A c-unit change multiplies the hazard by $e^{c\beta}$, so $e^{10\times 0.024} = e^{0.24} = 1.271$: a 27.1% higher churn rate per 10 euros.

Part 2 — a binary covariate. $e^{-0.693} = 0.500$. Onboarded customers churn at *half* the rate of comparable customers who did not onboard, at every tenure.

Part 3 — test and interval. $z = -0.693/0.210 = -3.30$, so $p \approx 0.001$: clear evidence. Build the interval on the β scale, *then* exponentiate: $-0.693 \pm 1.96(0.210) = [-1.105, -0.281]$, so the hazard-ratio interval is $[e^{-1.105}, e^{-0.281}] = [0.331, 0.755]$ — asymmetric around 0.500, as every hazard-ratio interval is.

Part 4 — combining covariates. Hazards multiply, so log hazards add; with the expensive, non-onboarded customer as the numerator, $e^{20(0.024)}/e^{-0.693} = e^{1.173} = 3.23$.

Common mistake

Averaging the two hazard ratios, or exponentiating after adding standard errors. Combine on the *log* scale, exponentiate once at the end.

Outline

Proportional hazards is an assumption, not a result

Two checks — one formal and one visual

- **Schoenfeld residuals.** If HR is constant, the residuals for covariate j show *no trend in time*. Test the correlation between the residuals and (a transform of) time: a small p-value is evidence *against* proportionality.
- **Complementary log-log plot.** Under PH the curves $\log(-\log \hat{S}(t))$ for two strata differ by the constant β, so they should look **parallel** — crossing or converging curves are a warning.

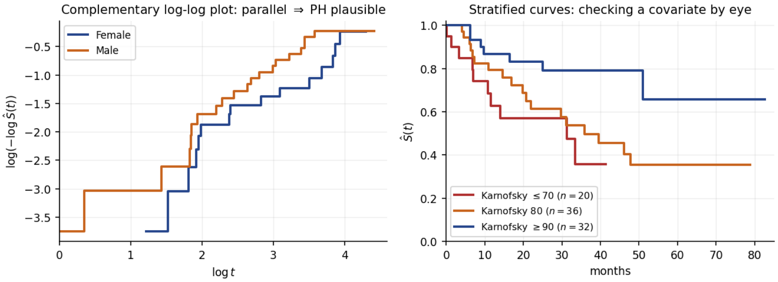
Schoenfeld test on all eight BrainCancer covariates

Every p-value exceeds 0.39 (`ki` 0.82, HG glioma 0.82, `gtv` 0.48; the smallest is 0.40, for `sex`). **No evidence against** proportional hazards — which with 35 events is a weak reassurance, not a clean bill of health.

Common mistake

Reading "$p = 0.82$" as "proportional hazards holds". Failing to reject is not accepting; with few events these tests have very little power.

Checking by eye



Takeaway

Left: the two $\log(-\log \hat{S})$ curves stay roughly parallel with a small constant gap, consistent with PH for sex. Right: stratifying by Karnofsky index gives three ordered, non-crossing curves ($n = 20, 36, 32$) — also what PH predicts. Neither plot proves the assumption; both would have shown a gross violation.

When proportional hazards fails: five honest options

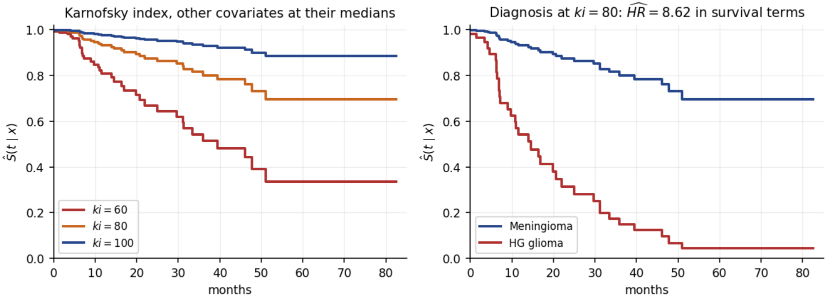
In rough order of how much you give up

1. **Stratify** on the offending variable: fit $h(t \mid x) = h_{0g}(t)e^{x^\top \beta}$ with a separate baseline per stratum. You lose its coefficient, and keep everything else.
2. **Interact with time**: let $\beta_j(t) = \beta_j + \gamma_j \log t$, and report an early and a late hazard ratio.
3. **Split the follow-up** into periods and fit a hazard ratio in each — the crude version of the same idea, and easy to explain.
4. **Change the target**: compare $\hat{S}$ at pre-specified horizons, or compare $\mathrm{RMST}(\tau)$. No proportionality needed at all.
5. **Use an accelerated-failure-time model** (Weibull, log-logistic), which multiplies *time* rather than the hazard.

Takeaway

A crossing-hazards problem is a *question* problem: "is A better than B?" has no time-free answer, so pick the horizon that matters to the decision and report the comparison there.

From coefficients to a curve for one subject



Takeaway

Combining $\hat{\beta}$ with Breslow's baseline gives $\hat{S}(t \mid x) = \hat{S}_0(t)^{\exp(x^\top \hat{\beta})}$. At 36 months, with the other covariates at their medians, $\hat{S} = 0.514$ for $\texttt{ki} = 60$ against 0.801 at 80 and 0.929 at 100; the median is 39.5 months at $\texttt{ki} = 60$ and is never reached above that. This is the form in which a clinician or a pricing team can use the model.

Evaluating a survival model

Harrell's concordance index

$$C = \Pr\left(\hat{\eta}_i > \hat{\eta}_j \;\middle|\; T_i < T_j\right), \qquad \hat{\eta}_i = x_i^\top \hat{\beta},$$

estimated over all *comparable* pairs — those whose ordering censoring lets us determine.

How to read this

- The survival analogue of the AUC: 0.5 is random ordering, 1.0 is perfect ranking. The BrainCancer fit reaches $C = 0.794$.
- It grades the *ranking* of risks only — a model can have $C = 0.79$ and still predict badly calibrated survival probabilities.
- So pair it with calibration — bucket subjects by $\hat{\eta}$ and overlay the predicted curve on each bucket's Kaplan–Meier curve — and cross-validate, exactly as in Chapter 5: read on the training data, C is optimistic.

In industry — Model risk: the two questions a validator asks

Discrimination — does it rank? (C-index, cross-validated.) *Calibration* — are the probabilities right at the horizons the business uses? A provisioning model fails validation on the second even when it passes the first.

High-dimensional survival: the regularised Cox model

Penalised partial likelihood

$$\max_{\beta} \left\{ \log \mathrm{PL}(\beta) \;-\; \lambda\Big[(1-\alpha)\tfrac{1}{2}\|\beta\|_2^2 + \alpha\|\beta\|_1\Big] \right\}$$

Chapter 6 transplanted

- With $p > n$ — 20 000 genes and 200 patients — the partial likelihood has no unique maximum and $\hat{\beta}$ does not exist. Exactly the Chapter 6 situation, one likelihood along.
- $\alpha = 1$ gives the **lasso**-Cox (a sparse risk score); $\alpha = 0$ the ridge-Cox; in between, the elastic net. Choose λ by cross-validating the partial likelihood, *not* the C-index alone.
- The resulting linear predictor $x^\top \hat{\beta}$ is the risk score behind commercial genomic assays.

In industry — Oncology diagnostics: from a signature to a decision

A lasso-Cox on gene expression yields a handful of genes and one score; the score is then dichotomised at a threshold chosen on *clinical* grounds and validated on an independent cohort. Sparsity is what makes the assay cheap enough to run.

Exercise 11.6 — Diagnosing and fixing a PH violation [Concept]

Exercise

A retention team fits a Cox model for time to churn. The Schoenfeld test for the covariate `annual_contract` returns $p = 0.002$; its complementary log-log curves cross at about month 14. The estimated hazard ratio is 0.98 with $p = 0.81$.

1. What is the substantive story behind crossing curves here?
2. Explain why $\mathrm{HR} = 0.98$, $p = 0.81$ is not evidence that contract type does not matter.
3. Propose two concrete fixes and state what each costs.
4. The team must publish a single number for the next board meeting. What do you recommend?

Solution — Exercise 11.6

Solution

Part 1 — the story. Annual contracts lock customers in, so their hazard is very low early and then spikes at renewal; monthly customers churn at a steady moderate rate. Before month 14 annual is protective, afterwards it is dangerous. That is a time-varying hazard ratio, i.e. a PH violation, and the crossing is not a nuisance — it *is* the finding.

Part 2 — why the null result is an artefact. A single β forces one constant hazard ratio, and the fit lands near the time-average of a protective-then-harmful effect. Early and late excesses cancel, so $\hat{\beta} \approx 0$ — the same cancellation that made the log-rank test blind to crossing curves. The model was asked the wrong question and answered it faithfully.

Part 3 — two fixes.

- **Split the follow-up** at month 14 and fit a hazard ratio in each period. Cost: the split point is a choice, and choosing it from these data invalidates the naive p-values.
- **Stratify** on `annual_contract`: every other coefficient stays interpretable and the baselines are free to differ. Cost: you get no hazard ratio for contract type at all.

Part 4 — the board number. Not a hazard ratio. Quote retention at a decision-relevant horizon — $\hat{S}(12)$ and $\hat{S}(24)$ by contract type with intervals — or $\mathrm{RMST}(24)$. Both are well defined when hazards cross.

Extended Exercise 11.2 — The Cox model in Python [Python]

Extended exercise (15 min)

Using `BrainCancer` and `lifelines`:

1. Fit a Cox model with all six covariates, coding `diagnosis` against **Meningioma**. Report $\hat{\beta}$, $e^{\hat{\beta}}$ and p for `ki` and for HG glioma.
2. Convert the `ki` coefficient into a hazard ratio for a ten-point increase, and write the one-sentence interpretation.
3. Report the concordance index, and the overall likelihood-ratio test.
4. Test proportional hazards with Schoenfeld residuals and state the conclusion.
5. Predict and plot $\hat{S}(t \mid x)$ for `ki` $\in \{60, 80, 100\}$ with the other covariates at their medians.

Run this live

The worked solution sits in `chapter_11_lab.ipynb` under *Lecture exercises — worked Python solutions*: the cell headed *Extended Exercise 11.2 — The Cox model in Python [Python]*.

Solution — Extended Exercise 11.2 (1/2)

Solution — code

```
import numpy as np, pandas as pd
from lifelines import CoxPHFitter
from lifelines.statistics import proportional_hazard_test
BC = load('BrainCancer', index_col=0).dropna() # 87 complete cases
BC['diagnosis'] = pd.Categorical(BC['diagnosis'], # Meningioma = reference
       categories=['Meningioma', 'HG glioma', 'LG glioma', 'Other'])
X = pd.get_dummies(BC, drop_first=True).astype(float) # drop the reference level
cph = CoxPHFitter().fit(X, duration_col='time', event_col='status')
print(cph.summary[['coef', 'exp(coef)', 'p']].round(4)) # log HR, HR, p-value
print('HR per 10 ki : %.3f' % np.exp(10 * cph.params_['ki'])) # rescaled HR
print('C-index : %.3f' % cph.concordance_index_) # discrimination
print(proportional_hazard_test(cph, X, time_transform='rank').summary['p'])
med = X.drop(columns=['time','status']).median().to_frame().T # median profile
prof = pd.concat([med] * 3, ignore_index=True) # three copies of it
prof['ki'] = [60., 80., 100.] # vary one covariate only
cph.predict_survival_function(prof).plot() # S(t | x) per profile
```

Solution — Extended Exercise 11.2 (2/2)

Solution — output and reading

- **(1)** `ki`: $\hat{\beta} = -0.0550$, $e^{\hat{\beta}} = 0.947$, $p = 0.003$. HG glioma: $\hat{\beta} = 2.1546$, $e^{\hat{\beta}} = 8.624$, $p < 0.001$. The remaining six covariates all have $p > 0.12$.
- **(2)** $e^{10\times(-0.0550)} = e^{-0.550} = 0.577$: *"holding tumour type, volume, site, sex and technique fixed, ten more Karnofsky points are associated with a 42% lower death rate at every follow-up time."*
- **(3)** $C = 0.794$ — good discrimination, but read on the training data, so optimistic. Likelihood-ratio test $\chi^2_8 = 41.4$, $p = 1.8 \times 10^{-6}$: the covariate block clearly carries signal.
- **(4)** Schoenfeld p-values run from 0.40 to 0.82: no evidence against proportional hazards. With 35 events, that is weak reassurance.
- **(5)** $\hat{S}(36 \mid \texttt{ki}) = 0.514, 0.801, 0.929$ for $\texttt{ki} = 60, 80, 100$. Median 39.5 months at 60; not reached at 80 or 100 within the 82.56 months of follow-up.

Common mistake: letting `get_dummies` pick the reference level alphabetically — here that is HG glioma, and every diagnosis coefficient changes sign and meaning while the fit stays identical.

Extended Exercise 11.3 — Publication bias, measured [Integrative]

Extended exercise (15 min)

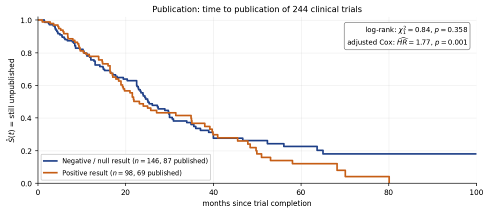
`Publication` records the time in months from completion to publication for 244 NHLBI-funded clinical trials; `status` $= 1$ means published, and `posres` $= 1$ means the trial found a statistically significant primary result.

1. A log-rank test comparing publication times by `posres` gives $\chi^2_1 = 0.84$, $p = 0.358$. What does that appear to say about publication bias?
2. A Cox model with `posres` alone gives $\widehat{\mathrm{HR}} = 1.16$, $p = 0.360$. Adding `impact`, `clinend`, `multi`, `sampsize` and `budget` moves it to $\widehat{\mathrm{HR}} = 1.77$, $p = 0.001$. Explain how one variable can go from null to clearly significant.
3. Which covariate does most of that work, and by what mechanism?
4. State the conclusion in one sentence a journal editor would understand.

Solution — Extended Exercise 11.3 (1/2)

Solution — what the unadjusted comparison hides

Part 1 — the apparent answer. Taken alone, $p = 0.358$ says trials with and without a positive result are published at indistinguishable rates: no evidence of publication bias. The two Kaplan–Meier curves are nearly on top of each other (medians 23.1 vs 25.1 months).



Part 2 — why adjustment changes everything. The log-rank test is *marginal*: it cannot hold anything fixed. `posres` is entangled with the journal-level variables — and the entanglement works in the opposite direction to the effect of interest, masking it. Conditioning on those variables removes the mask, and the hazard ratio for publication rises from 1.16 to 1.77 ($p = 0.001$).

Solution — Extended Exercise 11.3 (2/2)

Solution — mechanism and conclusion

Part 3 — the variable doing the work. `impact`, the journal impact factor, with $\hat{\beta} = 0.0583$ ($\mathrm{HR} = 1.060$ per point, $z = 8.7$, $p < 0.001$) — by far the strongest term in the model. High-impact journals are slow; positive trials aim at them. So a positive result buys a faster *decision* but a slower *venue*, and the two effects cancel in the raw comparison. `clinend` (a clinical endpoint) also matters, $\mathrm{HR} = 1.73$, $p = 0.037$; `multi`, `sampsize` and `budget` do not ($p \geq 0.075$).

Part 4 — the sentence. *"Among comparable trials — same target journal impact, same endpoint type, same size and budget — those with a positive primary result were published at 1.8 times the rate of those without ($p = 0.001$), so the raw publication times understate publication bias rather than refuting it."*

The transferable lesson. A log-rank test answers "are these two groups different?"; a Cox model answers "is this variable associated with the hazard, given the others?". When those disagree, the covariates — not the tests — are telling you something.

Common mistake

Concluding from $p = 0.358$ that publication bias is absent. An unadjusted comparison can be null because there is no effect, or because two real effects cancel; only the adjusted model distinguishes them.

Outline

Python lab — run it live in the notebook

Companion notebook: `chapter_11_lab.ipynb`

The code in this section is meant to be **demonstrated live**. Open the companion Jupyter notebook and run it cell by cell:

- §1, *Data and the censoring convention* — load `BrainCancer`, establish that `status` $= 1$ is death;
- §2, *Why the naive analyses fail* — reproduce the medians 11.6, 23.7, 47.8;
- §3, *Kaplan–Meier* — the curve, its confidence band and the curves by sex;
- §4, *Log-rank test*, then §5, *Cox proportional-hazards model*;
- §6, *Checking proportional hazards and predicting curves.*

Data loads via the `ISLP` package or the bundled CSV files; the notebook then closes with *Lecture exercises — worked Python solutions* and §7, *Exercises*.

Kaplan–Meier and the log-rank test in lifelines

```
import pandas as pd # data frames
from lifelines import KaplanMeierFitter # product-limit fitter
from lifelines.statistics import logrank_test # two-curve comparison

BC = pd.read_csv('BrainCancer.csv', index_col=0) # index col = row number
km = KaplanMeierFitter().fit(BC['time'], BC['status']) # status: 1 = death
print('median survival :', km.median_survival_time_) # 47.8 months

male = BC['sex'] == 'Male' # the two groups
res = logrank_test(BC.loc[male, 'time'], BC.loc[~male, 'time'],
                   BC.loc[male, 'status'], BC.loc[~male, 'status'])
print('chi2 = %.4f p = %.4f' % (res.test_statistic, res.p_value))
```

Takeaway

Prints `median survival : 47.8` and `chi2 = 1.4405 p = 0.2301` — the numbers quoted throughout this deck.

The Cox model and its diagnostics

```
import numpy as np, pandas as pd
from lifelines import CoxPHFitter # partial-likelihood fit
from lifelines.statistics import proportional_hazard_test

BC = pd.read_csv('BrainCancer.csv', index_col=0).dropna() # 87 complete cases
BC['diagnosis'] = pd.Categorical(BC['diagnosis'], # set the reference
       categories=['Meningioma', 'HG glioma', 'LG glioma', 'Other'])
X = pd.get_dummies(BC, drop_first=True).astype(float) # design matrix

cph = CoxPHFitter().fit(X, duration_col='time', event_col='status')
cph.print_summary(columns=['coef', 'exp(coef)', 'p']) # log HR and HR
print('HR per 10 ki : %.3f' % np.exp(10 * cph.params_['ki'])) # 0.577
print('C-index : %.3f' % cph.concordance_index_) # 0.794
proportional_hazard_test(cph, X, time_transform='rank').print_summary()
```

Takeaway

print_summary gives $\hat{\beta}$, $e^{\hat{\beta}}$, its interval and p; proportional_hazard_test gives one p-value per covariate. Fit, read, *then* check — in that order, every time.

Outline

Chapter 11 in one slide

What we accomplished

Learning from outcomes that are times, when for many subjects the clock is still running.

- Right-censoring: the observed pair (Y, δ), and the independent-censoring assumption everything rests on.
- The survival curve $S(t)$ and its non-parametric Kaplan–Meier estimate, with a confidence band.
- The log-rank test for two curves — and the crossing-hazards case it cannot see.
- The Cox proportional-hazards model: covariates through the hazard, β estimated by a partial likelihood in which the baseline cancels.
- Coefficients as log hazard ratios; checking proportional hazards; the C-index; regularised Cox when $p > n$.

Key formulas at a glance

Observed data	$Y_i = \min(T_i, C_i)$, $\delta_i = 1\{T_i \le C_i\}$; independent censoring $T_i \perp C_i \mid x_i$.
Survival function	$S(t) = \Pr(T > t)$; median $= \min\{t : S(t) \le 0.5\}$; $E[T] = \int_0^\infty S(t)dt$; $\mathrm{RMST}(\tau) = \int_0^\tau S(t)dt$.
Hazard	$h(t) = \lim_{\Delta\to 0} \Pr(t < T \le t+\Delta \mid T > t)/\Delta = -\frac{d}{dt}\log S(t)$; $H(t) = \int_0^t h(u)du$; $S(t) = e^{-H(t)}$.
Kaplan–Meier	$\hat{S}(t) = \prod_{k:t_k \le t}(1 - d_k/r_k)$, with $r_k = \#\{i : Y_i \ge t_k\}$.
Greenwood	$\widehat{\mathrm{Var}}(\hat{S}(t)) \approx \hat{S}(t)^2 \sum_{k:t_k\le t} \dfrac{d_k}{r_k(r_k - d_k)}$.
Log-rank	$E_{1k} = d_k \dfrac{r_{1k}}{r_k}$, $V_{1k} = \dfrac{d_k\,(r_{1k}/r_k)(1 - r_{1k}/r_k)(r_k - d_k)}{r_k - 1}$, $Z^2 = \dfrac{(O_1 - E_1)^2}{V_1} \sim \chi^2_1$.
Cox model	$h(t \mid x) = h_0(t)\exp(x^\top\beta)$; $S(t \mid x) = S_0(t)^{\exp(x^\top\beta)}$.
Partial likelihood	$\mathrm{PL}(\beta) = \prod_{i:\delta_i=1} \dfrac{\exp(x_i^\top\beta)}{\sum_{j\in R(Y_i)}\exp(x_j^\top\beta)}$, $R(t) = \{j : Y_j \ge t\}$.
Hazard ratio	$\mathrm{HR}_j = e^{\beta_j}$; a c-unit change gives $e^{c\beta_j}$; $\mathrm{CI} = \exp(\hat\beta_j \pm 1.96\,\mathrm{se}(\hat\beta_j))$; $z = \hat\beta_j/\mathrm{se}(\hat\beta_j)$.
C-index	$C = \Pr(\hat\eta_i > \hat\eta_j \mid T_i < T_j)$ over comparable pairs, $\hat\eta_i = x_i^\top\hat\beta$.
Regularised Cox	$\max_\beta\{\log\mathrm{PL}(\beta) - \lambda[(1-\alpha)\frac{1}{2}\|\beta\|_2^2 + \alpha\|\beta\|_1]\}$.

Vocabulary checklist

Term	Meaning
Right-censoring	Event not yet observed; all we know is $T > Y$
Event indicator δ	$1 =$ event seen, $0 =$ censored
Independent censoring	$T \perp C \mid x$; censored subjects are representative
Administrative censoring	Follow-up ended by the calendar, not by the subject
Risk set $R(t)$	Subjects with $Y_i \geq t$: still alive and still watched
Survival function $S(t)$	$\Pr(T > t)$
Hazard $h(t)$	Instantaneous event rate among those at risk
Cumulative hazard $H(t)$	$\int_0^t h$; $S = e^{-H}$
Kaplan–Meier	Product-limit estimator of S
Greenwood's formula	Variance of $\hat{S}(t)$, hence the confidence band
RMST(τ)	Restricted mean survival: $\int_0^\tau S$
Log-rank test	χ^2_1 comparison of two whole curves
Proportional hazards	Hazard ratio constant in t — an assumption
Partial likelihood	Cox's likelihood in which $h_0(t)$ cancels
Hazard ratio e^{β_j}	Multiplicative effect on the rate, not on the risk
Schoenfeld residuals	Diagnostic for a time-varying hazard ratio
C-index	Concordance: the survival analogue of AUC

Decision rules of thumb

- Check the coding of the event indicator *against the data* before you fit anything; state the convention in the write-up.
- Plot Kaplan–Meier curves, with censoring marks, before any test or model.
- One binary grouping variable $\Rightarrow$ log-rank. Several covariates, or a continuous one $\Rightarrow$ Cox.
- Never report a mean survival time from censored data; report a median, a probability at a stated horizon, or RMST with its τ.
- Report the number of **events**, not only the sample size — it is what the precision depends on.
- Check proportional hazards; if it fails, change the target rather than forcing one hazard ratio.
- $p > n \Rightarrow$ regularise the partial likelihood, and pick λ by cross-validation.

Common pitfalls

Don't

1. Drop censored subjects, or treat their last-seen time as an event time.
2. Get the event indicator backwards — every curve inverts and nothing warns you.
3. Read a censoring mark as a step down in $\hat{S}$.
4. Report a median survival that the curve never reaches.
5. Call a hazard ratio a risk ratio, an odds ratio, or a ratio of survival times.
6. Quote a hazard ratio without its reference level and its units.
7. Treat proportional hazards as established because a test failed to reject it.
8. Conclude "no difference" from a large log-rank p-value when the curves cross.
9. Extrapolate $\hat{S}$ beyond the largest observed follow-up time.
10. Assume censoring is independent when subjects withdrew because they were deteriorating.

Self-check questions

Quick self-assessment

1. A patient is censored at 30 months. What exactly does that record contribute to $\hat{S}(40)$?
2. Why does $\hat{S}$ not step down at a censoring time?
3. Two curves cross and the log-rank p-value is 0.6. What do you report?
4. In $h(t \mid x) = h_0(t)e^{x^\top \beta}$, which part is estimated by the partial likelihood, and which is not?
5. $\hat{\beta} = 0.4$ for a binary covariate. Give the hazard ratio and one sentence of interpretation.

Answers

(1) It keeps that patient in every risk set r_k for $t_k \leq 30$, which lowers every later r_k by one — so it changes the size of subsequent steps, not their location. (2) Nobody died: the factor $1 - d_k/r_k$ is only formed at event times. (3) Not "no difference" — report $\hat{S}$ at pre-specified horizons, or RMST. (4) β is; $h_0(t)$ is not — it cancels, and needs Breslow's estimator if you want absolute risk. (5) $e^{0.4} = 1.49$: a 49% higher event rate at every time, holding the other covariates fixed.

Five things to remember

Chapter 11 in five lines

1. A censored subject is **information**, not a missing row: it belongs in every risk set up to the moment it leaves.
2. Independent censoring is the assumption that makes all of this work — argue for it from the study design, because no test can check it.
3. Kaplan–Meier is arithmetic on at-risk sets; the log-rank test is observed-minus-expected over those same sets.
4. Cox estimates β without ever modelling the baseline hazard, and $e^{\hat{\beta}_j}$ is a hazard ratio — a ratio of rates, not of risks.
5. Proportional hazards is an assumption. Check it; and when it fails, change the question rather than the p-value.

What's next

Where this chapter connects

- **Back to Chapter 6:** the regularised partial likelihood is the lasso and ridge machinery, applied to a different loss.
- **Back to Chapter 5:** cross-validate the C-index — an in-sample 0.794 is optimistic for exactly the Chapter 5 reasons.
- **Chapter 12:** unsupervised learning, where there is no response at all.
- **Chapter 13:** multiple testing — and a survival study with many subgroups is precisely a multiplicity problem.

Takeaway

Chapters 11 and 12 are the two "unusual response" chapters: here the response is only partly observed, there it is absent. Both are handled by changing the likelihood, not the loss you happen to know.

Appendix — what is in here, and why

Nothing in the appendix is needed to follow the chapter: each slide is a formal derivation, a longer worked exercise, or a side topic. Read it when you want the full story, or when a later chapter sends you back here.

Contents of the appendix

- **Greenwood's formula** — where the confidence band comes from and why it widens; the main deck only needs to *read* the band.
- **Other censoring and truncation** — left-, interval- and left-truncated data. Real in practice, but off the chapter's arc.
- **Deriving the partial likelihood** — the risk-set argument in full, plus how ties are handled.
- **The log-rank test is a score test** — the algebraic link between the chapter's two headline methods, with the BrainCancer numbers.
- **Time-varying covariates and immortal-time bias** — the most common way a real survival analysis is quietly wrong.
- **Exercise 11.7** — an exponential-model drill a 180-minute session usually cannot fit.

Greenwood's formula

Variance of the Kaplan–Meier estimator

$$\widehat{\mathrm{Var}}(\hat{S}(t)) \approx \hat{S}(t)^2 \sum_{k:\, t_k \le t} \frac{d_k}{r_k\,(r_k - d_k)}.$$

Sketch and how to use it

- Write $\log \hat{S}(t) = \sum_k \log(1 - d_k/r_k)$. Treat the d_k as independent binomials given r_k, so $\widehat{\mathrm{Var}}(d_k/r_k) = \frac{(d_k/r_k)(1-d_k/r_k)}{r_k}$, propagate through the logarithm, and sum.
- Multiplying back by $\hat{S}(t)^2$ (the delta method) gives the formula.
- Each term grows as r_k shrinks: that is precisely why the band widens to the right.
- Software reports a *log–log-transformed* interval, so the endpoints stay inside $[0, 1]$ and the interval is asymmetric.

BrainCancer at 20 months

$\hat{S}(20) = 0.7132$ and $\sum d_k/(r_k(r_k - d_k)) = 0.005101$, so $\mathrm{se} = 0.7132\sqrt{0.005101} = 0.0509$. The naive interval $0.7132 \pm 1.96(0.0509) = [0.613, 0.813]$; `lifelines`' log–log interval is $[0.600, 0.800]$ — same information, better behaved near the boundaries.

Other censoring, and truncation

Four situations that are not right-censoring

- **Left-censoring:** the event had already happened before observation began — we know $T < Y$. Common in infection studies where a subject is already seropositive at enrolment.
- **Interval-censoring:** the event happened between two visits, so $T \in (a, b]$. Any periodically screened cohort produces this; treating the visit date as the event date biases times upwards.
- **Left-truncation (delayed entry):** a subject only enters the study after surviving to age a, so must be excluded from the risk set before a. Ignoring it makes survival look better than it is.
- **Competing risks:** death from another cause makes the event of interest unobservable forever. Kaplan–Meier on the cause of interest then *overstates* incidence; use a cumulative-incidence function instead.

Takeaway

Right-censoring is the easy case, and the one the chapter treats. The others need different likelihoods — but the same discipline of asking exactly what each record proves.

Deriving the partial likelihood, and ties

The risk-set argument

Condition on the fact that *some* subject in $R(t_k)$ died at t_k, and ask which one. Under the Cox model the instantaneous rate for subject j is $h_0(t_k)e^{x_j^\top \beta}$, so

$$\Pr(\text{subject } i \text{ is the one}) = \frac{h_0(t_k)e^{x_i^\top \beta}}{\sum_{j \in R(t_k)} h_0(t_k)e^{x_j^\top \beta}} = \frac{e^{x_i^\top \beta}}{\sum_{j \in R(t_k)} e^{x_j^\top \beta}}.$$

The nuisance $h_0(t_k)$ cancels because it is common to every subject at risk at that instant. Multiplying these conditional probabilities over all observed events gives $\mathrm{PL}(\beta)$; it is not a likelihood for the data, but it behaves like one.

Ties

With continuous time, ties have probability zero; with monthly or daily data they are everywhere. **Breslow**: use the same denominator for all tied events (fast, biased towards 0 when ties are heavy). **Efron**: discount the denominator progressively across the tied set (the `lifelines` default, and nearly always the better choice). **Exact**: average over all orderings — correct and expensive.

The log-rank test is a score test of the Cox model

Why the chapter's two headline methods are one

Fit a Cox model with a single binary covariate $x_i = 1\{i \in \text{group } 1\}$. The score of the partial log-likelihood at $\beta = 0$ is

$$\left.\frac{\partial \log \mathrm{PL}}{\partial \beta}\right|_{\beta=0} = \sum_k \left(d_{1k} - d_k \frac{r_{1k}}{r_k}\right) = O_1 - E_1,$$

and the corresponding information is V_1 (with Breslow's tie handling). So the score test $(O_1 - E_1)^2/V_1$ **is** the log-rank statistic.

BrainCancer split by sex

Log-rank: $\chi^2_1 = 1.4405$, $p = 0.2301$. Cox with sex alone: $\hat{\beta} = 0.4077$ $(\mathrm{HR} = 1.503)$, Wald $z = 1.192$ giving $p = 0.2333$, and a likelihood-ratio test of 1.4388, $p = 0.2303$. Three asymptotically equivalent tests of the same null, agreeing to two decimals.

Takeaway

The log-rank test is the assumption-light special case; Cox is the general machine. That is also why the log-rank test inherits Cox's blind spot for crossing hazards.

Time-varying covariates and immortal-time bias

The setup

Some covariates change during follow-up: a patient starts a drug at month 7, a customer upgrades at month 14. The Cox model handles this by letting $x_i(t)$ enter the risk set as it is *at that instant*: $h(t \mid x_i(t)) = h_0(t)\exp\{x_i(t)^\top \beta\}$, fitted from a data set split into one row per subject per interval.

Immortal-time bias

Classifying a subject as "treated" from time zero when treatment only began at month 7 grants them seven months during which they *could not* have been classified as an untreated death. Treatment then looks protective purely by construction — an error that has produced real and repeatedly retracted findings on statins, screening and adherence.

Takeaway

A covariate may only use information available at the time it is applied. If the exposure has a start date, the analysis must have one too.

Exercise 11.7 — The exponential model [Math]

Exercise

Suppose T has a constant hazard, $h(t) = \lambda$ for all $t > 0$.

1. Derive $H(t)$ and $S(t)$.
2. Find the median survival time in terms of λ.
3. Show that this family satisfies proportional hazards when $\lambda_i = \lambda_0 e^{x_i^\top \beta}$, and identify the baseline hazard.
4. With n subjects, $D = \sum_i \delta_i$ events and total follow-up $\sum_i Y_i$, the maximum-likelihood estimator is $\hat{\lambda} = D / \sum_i Y_i$. Evaluate it for BrainCancer ($D = 35$, $\sum_i Y_i = 2416.26$ months) and compare the implied median with the Kaplan–Meier median of 47.8.

Solution — Exercise 11.7

Solution

Part 1. $H(t) = \int_0^t \lambda\, du = \lambda t$, so $S(t) = e^{-H(t)} = e^{-\lambda t}$: the exponential distribution, the unique memoryless survival law.

Part 2. Solve $e^{-\lambda t} = 0.5$: $t_{0.5} = \dfrac{\log 2}{\lambda} = \dfrac{0.6931}{\lambda}$.

Part 3. With $\lambda_i = \lambda_0 e^{x_i^\top \beta}$ the hazard is $h(t \mid x_i) = \lambda_0 e^{x_i^\top \beta}$, which is exactly $h_0(t) e^{x_i^\top \beta}$ with the *constant* baseline $h_0(t) \equiv \lambda_0$. The ratio of two hazards is $e^{(x_i - x_j)^\top \beta}$, free of t: an exponential model is a Cox model with the baseline pinned down.

Part 4. $\hat{\lambda} = 35/2416.26 = 0.014485$ deaths per month of follow-up, so the implied median is $\log 2/0.014485 = 47.85$ months — strikingly close to the Kaplan–Meier value of 47.8. Reassuring, but partly luck: the constant-hazard assumption spans nearly seven years, and the two estimates diverge at other quantiles.

Common mistake

Estimating λ as D/n or as $1/\bar{Y}$ over events only. The denominator must be **total time at risk**, $\sum_i Y_i$, which is what censored subjects contribute.